

A Practical Guide to Federated learning

TOPIC -- FEDERATED LEARNING

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Covariate shift: local nodes may store examples that have different statistical distributions compared to other nodes. An example occurs in natural language processing datasets where people typically write the same digits/letters with different stroke widths or slants. Prior probability shift: local nodes may store labels that have different statistical distributions compared to other nodes. This can happen if datasets are regional and/or demographically partitioned. For example, datasets containing images of animals vary significantly from country to country. Concept drift (same label, different features): local nodes may share the same labels but some of them correspond to different features at different local nodes. For example, images that depict a particular object can vary according to the weather condition in which they were captured. Concept shift (same features, different labels): local nodes may share the same features but some of them correspond to different labels at different local nodes. For example, in natural language processing, the sentiment analysis may yield different sentiments even if the same text is observed. Unbalanced: the amount of data available at the local nodes may vary significantly in size. The loss in accuracy due to non-iid data can be bounded through using more sophisticated means of doing data normalization, rather than batch normalization.

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Federated stochastic gradient descent (FedSGD) Stochastic gradient descent is an approach used in deep learning, where gradients are computed on a random subset of the total dataset and then used to make one step of the gradient descent.. Federated stochastic gradient descent is the analog of this algorithm to the federated setting, but uses a random subset of the nodes, each node using all its data. The server averages the gradients in proportion to the number of training data on each node, and uses the average to make a gradient descent step.

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Federated Proximal (FedProx) Federated Proximal (FedProx) extends FedAvg by adding a proximal term to the local objective in the form of $\frac{\mu}{2} \|w - w_t\|^2$, where μ is the FedProx hyper parameter, in order to address data heterogeneity across clients.

The proximal term constrains local updates, helping reduce client drift when client data are non-IID.

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Federated learning (also known as collaborative learning) is a machine learning technique in a setting where multiple entities (often called clients) collaboratively train a model while keeping their data decentralized, rather than centrally stored. A defining characteristic of federated learning is data heterogeneity. Because client data is decentralized, data samples held by each client may not be independently and identically distributed. Federated learning is generally concerned with and motivated by issues such as data privacy, data minimization, and data access rights. Its applications involve a variety of research areas including defence, telecommunications, the Internet of things, and pharmaceuticals.

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Biometrics Federated Learning (FL) is transforming biometric recognition by enabling collaborative model training across distributed data sources while preserving privacy. By eliminating the need to share sensitive biometric templates like fingerprints, facial images, and iris scans, FL addresses privacy concerns and regulatory constraints, allowing for improved model accuracy and generalizability. It mitigates challenges of data fragmentation by leveraging scattered datasets, making it particularly effective for diverse biometric applications such as facial and iris recognition. However, FL faces challenges, including model and data heterogeneity, computational overhead, and vulnerability to security threats like inference attacks. Future directions include developing personalized FL frameworks, enhancing system efficiency, and expanding FL applications to biometric presentation attack detection (PAD) and quality assessment, fostering innovation and robust solutions in privacy-sensitive environments.

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Those parameters have to be optimized depending on the constraints of the machine learning application (e.g., available computing power, available memory, bandwidth). For instance, stochastically choosing a limited fraction C of nodes for each iteration diminishes computing cost and may prevent overfitting, in the same way that stochastic gradient descent can reduce overfitting.

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A further governance-related issue concerns the distribution of value among participants as multi-party federated learning systems scale. As more clients contribute to training, model accuracy may approach a performance threshold, after which additional contributions yield diminishing marginal utility. This dynamic can create tensions within collaborative arrangements, as early participants may perceive later contributions as offering reduced value while still benefiting from the shared model. Such asymmetries may affect incentives to join or remain in a consortium and complicate the design of fair participation, ownership, or reward structures. Establishing transparent rules for contribution assessment and benefit allocation is therefore an important organisational challenge and represents a direction for future research.

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Decentralized federated learning In the decentralized federated learning setting, the nodes are able to coordinate themselves to obtain the global model. This setup prevents single point failures as the model updates are exchanged only between interconnected nodes without the orchestration of the central server. Nevertheless, the specific network topology may affect the performances of the learning process. See blockchain-based federated learning and the references therein.

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HyFDCA is a provably convergent primal-dual algorithm for hybrid FL in at least the following settings. **Hybrid Federated Setting with Complete Client Participation** Horizontal Federated Setting with Random Subsets of Available Clients The authors show HyFDCA enjoys a convergence rate of $O(1/t)$ which matches the convergence rate of FedAvg (see below). **Vertical Federated Setting with Incomplete Client Participation** The authors show HyFDCA enjoys a convergence rate of $O(\log(t)t)$ whereas FedBCD exhibits a slower $O(1/\sqrt{t})$ convergence rate and requires full client participation. HyFDCA provides the privacy steps that ensure privacy of client data in the primal-dual setting. These principles apply to future efforts in developing primal-dual algorithms for FL. HyFDCA empirically outperforms HyFEM and FedAvg in loss function value and validation accuracy across a multitude of problem settings and datasets (see below for more details). The authors also introduce a hyperparameter selection framework for FL with competing metrics using ideas from multiobjective optimization. There is only one other algorithm that focuses on hybrid FL, HyFEM proposed by Zhang et al. (2020). This algorithm uses a feature matching formulation that balances clients building accurate local models and the server learning an accurate global model. This requires a matching regularizer constant

that must be tuned based on user goals and results in disparate local and global models. Furthermore, the convergence results provided for HyFEM only prove convergence of the matching formulation not of the original global problem. This work is substantially different than HyFDCA's approach which uses data on local clients to build a global model that converges to the same solution as if the model was trained centrally. Furthermore, the local and global models are synchronized and do not require the adjustment of a matching parameter between local and global models. However, HyFEM is suitable for a vast array of architectures including deep learning architectures, whereas HyFDCA is designed for convex problems like logistic regression and support vector machines. HyFDCA is empirically benchmarked against the aforementioned HyFEM as well as the popular FedAvg in solving convex problem (specifically classification problems) for several popular datasets (MNIST, Covtype, and News20). The authors found HyFDCA converges to a lower loss value and higher validation accuracy in less overall time in 33 of 36 comparisons examined and 36 of 36 comparisons examined with respect to the number of outer iterations. Lastly, HyFDCA only requires tuning of one hyperparameter, the number of inner iterations, as opposed to FedAvg (which requires tuning three) or HyFEM (which requires tuning four). In addition to FedAvg and HyFEM being quite difficult to optimize hyperparameters in turn greatly affecting convergence, HyFDCA's single hyperparameter allows for simpler practical implementations and hyperparameter selection methodologies.